Leg Design for a Humanoid Walking Robot

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***Abstract*— The paper presents the leg design of the 22-DoF humanoid walking robot *Lola*. The goal of the project is to realize a fast, human-like walking motion. The robot is characterized by its lightweight construction, a modular, multi-sensory joint design with brushless motors and an electronics architecture using decentralized joint controllers and sensor data processing. Linear actuators are used for the knee and ankle joints to achieve a better mass distribution in the legs and to improve performance of the stabilizing control. Some critical structural parts have been designed by means of topology optimization in order to balance the demands for high stiffness and strength with low weight.**

I. Introduction

Due to recent developments in enabling technologies (biped walking control, mechatronics, computer technology) and findings in other fields like biomechanics, the development of humanoid robots is currently in an exciting phase. Prominent examples of sophisticated robots are Honda's *Asimo* [1], *H7* from Tokyo University [2], *HRP-2* from AIST and Kawada Industries [3], the world's first running biped *QRIO* from Sony [4], *Wabian-2* from Waseda University [5] and the Korean robot *HUBO* from KAIST [6].

Early research on humanoid robot action planning and control mainly focused on biped locomotion. Even if most robots achieve reliable dynamic walking, compared with human beings, fast walking is still a challenge. Research in this area is continuing, since there still are many unsolved problems, e.g. fast walking and running [7], [8], sudden turning motions, walking on rough terrain and trajectory generation in complex environments. Currently, research on so-called "whole body motion" of humanoid robots is increasing. Even though walking also involves motion of the whole body, new problems arise in tasks like carrying or manipulating objects while walking. Therefore, the goal is to fully utilize the wide range of capabilities inherent in humanoid robots in order to achieve goals, like walking, crawling, stair climbing and reaching.

Results from the *Johnnie* project [9], where a maximum speed of 2.4 [km/h] has been achieved, suggest that significant increases in walking speed, dexterity and autonomy are only possible if the robot's hardware is updated to state-of-the-art mechatronics technology. Currently the humanoid walking robot *Lola* (Fig. 1) with enhanced performance is being developed. The aim is to realize a fast, human-like walking motion, including a significant increase in walking speed, more flexible gait patterns and increased autonomy.

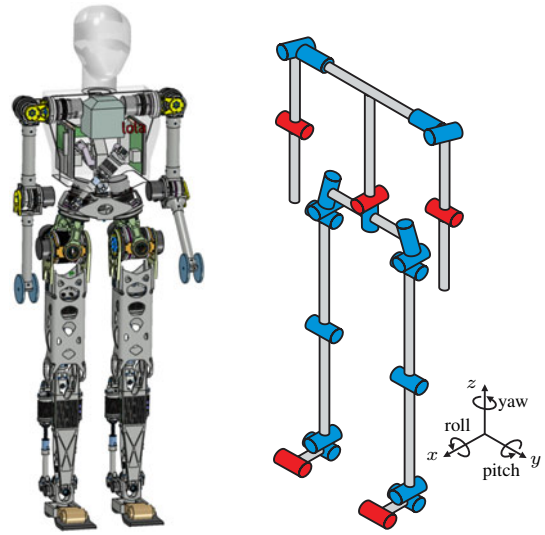


Fig. 1. CAD model (left) and joint structure of *Lola* (right). Newly introduced DoF are marked in red.

One of the keys to faster walking is greater robustness and stability. The new control architecture tries to achieve this by adding an on-line adaptation of gait parameters such as step length and width inspired by biomechanical findings. But fast biped walking also requires robust, reliable and high-performance hardware. Especially the legs and feet of the robot must be designed carefully in order to achieve good dynamic behavior. Since the robot's weight has a strong influence on global system dynamics, the lightweight construction is of great importance and must be balanced with the demand for high stiffness and powerful drives necessary in order to achieve the desired torques and speeds at the required bandwidths.

The paper is organized as follows. Section II gives an overview of the robot. In section III the design of structural components aided by topology optimization is presented, and section IV comments on the selection of appropriate actuation kinematics, especially of multi-DoF (degree of freedom) joints. Section V concludes with the mechanical design of the legs.

II. Specifications of Humanoid Robot *Lola*

A. Kinematic Structure

Fast locomotion poses a significant challenge and requires an adequate design of the overall mechatronic system. One

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of the most important questions regarding hardware design is how to choose the robot's kinematic structure in order to enable a natural, stable and fast gait. Experiments and simulations have shown that additional, redundant DoF can increase the robot's range of motion, flexibility and stability of gait patterns and the walking speed. Based on results from biomechanical research and experience with *Johnnie* [9] we chose a configuration with 22 actively driven DoF for *Lola*: The legs have 7 DoF each, while the upper body has two and each arm has three. Fig. 1 shows a CAD model of the robot and a schematic representation of its joint structure, where DoF newly introduced in *Lola* are shown in red. The physical dimensions are based on anthropometric data and correspond with an 180 [cm] tall adult.

The 2-DoF waist allows torso and pelvis to roll and yaw independently in order to increase step length and to provide more mobility for lateral motions. For example the roll axis facilitates walking with a straight stance leg at nearly constant height of the center of gravity [10]. The waist joint is driven by a parallel mechanism with ballscrews. The configuration is similar to the ankle joint (cf. V-C).

Nearly all humanoid robots have 6-DoF legs and feet that consist of one rigid body. However, it is difficult to perform forward roll across the forefoot with such a one-piece foot. Even small disturbances would lead to instabilities due to the line contact between the foot's leading edge and the floor. Moreover, forward roll across the forefoot is expected to reduce the joint loads in hip and knee. Similar to the robots *H6* and *H7* [11] we propose actively driven toe joints for *Lola*.

Another speciality in the leg structure is the knee joint which is actuated by a linear drive based on a ballscrew. The torque-speed characteristic of this actuator corresponds to the human knee, and despite the high performance the inertia of the thigh remains acceptable.

B. Modular Joint Concept

In most humanoid robots, the predominant actuation principle is the combination of Harmonic Drive gears with brush DC motors or, rarely, brushless DC motors [2], [3], [5], [6]. Timing belts are used in many joints as additional speed reducers and to improve mass distribution, e.g. by shifting the knee motors towards the hip.

Lola is characterized by a modular, multi-sensory joint design. This concept consists of highly integrated, mechatronic joint modules where motor, gear and joint sensors are merged into a compact drive unit. High performance AC permanent magnet servomotors are used because of their superior power density. We use frameless motors from Parker Bayside which allow an integrated design optimized for small space and low weight. Linear drives based on ballscrews are used in the waist, knee and ankle joints, all other joints employ Harmonic Drive gears. An absolute angular encoder as link position sensor compensates elasticities and nonlinearities in the drive chain and eliminates the need for a homing routine, making startup faster and easier. Details on the joint concept can be found in [12].

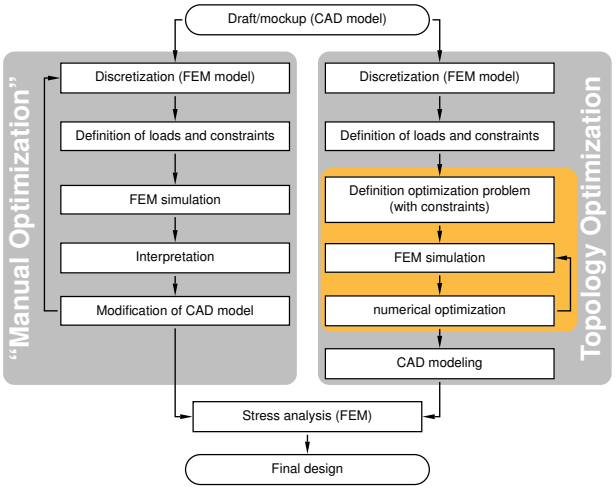


Fig. 2. Comparison of the workflow of conventional part design and the automated process based on topology optimization

C. Electronics Architecture

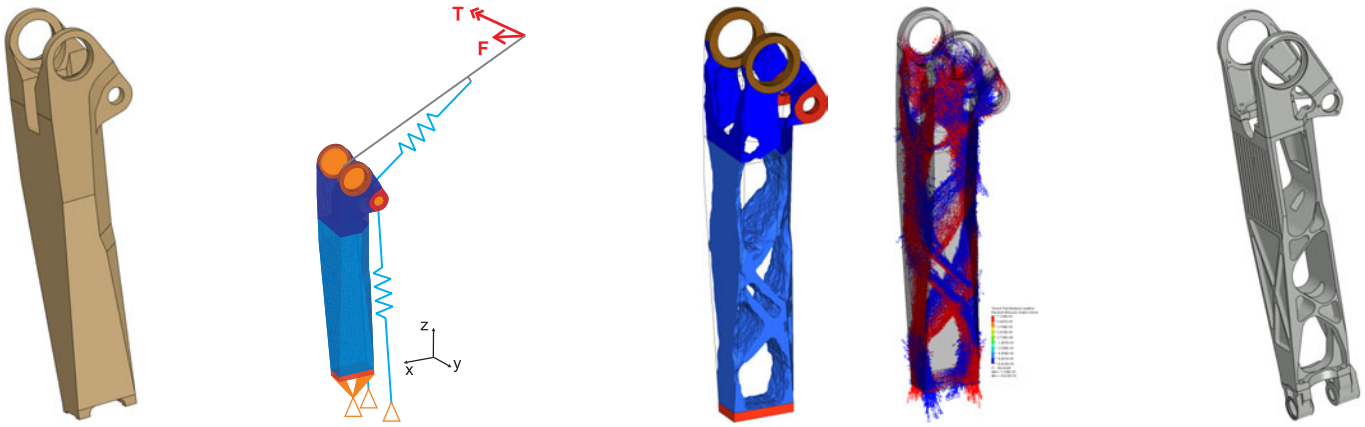
Besides the modular joint concept, the informational structure of the robot as "intelligent sensor-actuator network" accounts for the reduction of overall complexity [12]. Decentral control of global system dynamics is not desirable due to the highly coupled kinematics and dynamics. Control algorithms for stabilizing global dynamics are therefore processed by a central controller instance. However, low-level control of link positions and velocities is shifted into local controllers. Sensor data processing of the force-torque sensors and the inertial measurement unit is also done locally. In a sense of informational pre-filtering only relevant data are transmitted to the higher-ranking control layer. *SERCOS III* is employed for communication between central unit, sensors and joint controllers [13]. It is the third generation of a real-time communication system, which is widely used in tooling and paper machines. It uses Ethernet physics and was specially developed for distributed motion control applications. *SERCOS III* is characterized by a high level of determinism, a guaranteed bandwidth and a minimum of protocol overhead.

III. Designing Structural Components Using Topology Optimization

The mechanical structure of the robot should show high strength and stiffness to prevent distortion, at the same time the overall weight has to be kept at a minimum. All highly loaded parts are analyzed with the finite element method (FEM), where the loads acting are calculated with the multibody simulation of the robot [14].

Most parts can be designed for low weight and ideal flux of force using only intuition and experience. This method of empirical part design ("manual optimization", Fig. 2 left) requires building and calculating a new FEM model each time the part geometry is modified. This task is simplified by many current CAD systems which incorporate FEM modules. In an

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(a) Mockup (b) Optimization model (c) Optimization result (d) Part design

Fig. 3. Topology optimization of the shank

iterative procedure the results are analyzed and the geometry is modified.

However, if a component has a complex multi-axial stress condition and strict geometric constraints, a solution that satisfies all requirements is difficult to find with this procedure. Here the topology optimization method (Fig. 2 right) has its strengths and aids the development of optimal part geometries regarding high stiffness, strength and low weight. This FEM-based method generates optimal concept design proposals from supplied packaging information and meets weight or stiffness targets and constraint criteria such as maximum allowable deflection, stress and method of manufacture. The workflow is identical for the first steps, in addition an optimization problem is formulated. The part geometry is determined iteratively without user interaction, using a numerical optimization method. The result is optimal in a mathematical sense and is superior to conventional designs if the optimization problem is set up properly. However, the implementation and verification of an optimization model is a very time consuming task, because constraints (e.g. bearings) considerably determine the optimization result. Also, the transfer of the design proposal into the actual part geometry is costly since special attention has to be paid to manufacturability and ease of assembly. Therefore the use of this tool must be carefully balanced in consideration of cost and benefit. For *Lola*, we designed pelvis, thigh and shank with *OptiStruct* from *Altair Engineering*, which is able to take manufacturing and symmetry constraints into account. It is also possible to reserve dedicated areas that must not be modified, e.g. screwings or flanges.

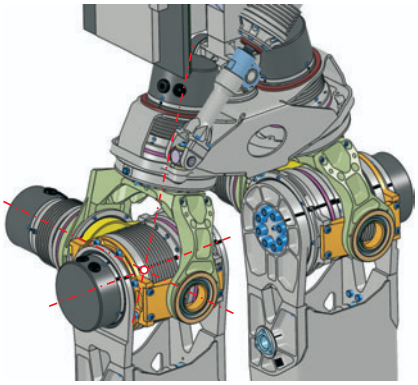
The individual steps of the topology optimization process are shown in Fig. 3 for the shank. This part connects the 1-DoF knee joint and the 2-DoF ankle joint that are both actuated by ballscrew drives. Therefore loads are transmitted not only at the joint flanges, but also at the hinging points of the drives. The designed space is mainly determined by the workspace of the ballscrew drives and the geometry of other leg segments. In a first step, a CAD mockup (Fig. 3(a)) which resembles the maximum allowable designed space of the part, is created by collision analysis. The mockup should already consider the intended manufacturing process and capabilities. Based on this mockup, the optimization model is built (Fig. 3(b)). Realistic results can only be achieved if the force transmission by the ballscrew drives is considered. Therefore the shank and the ballscrew drives for knee and ankle are modeled as elastic bending beams. The constraining forces which are obtained from the multibody simulation of the robot [14] are applied to the hip joint, the ankle joint is cardanically supported. The optimization result is shown in Fig. 3(c). The goal was to find a design proposal which can be manufactured by milling. The main tension in the righthand plot illustrates the optimized load spread and distribution in the shank. The last step is the final part design based on the optimization result as shown in Fig. 3(d).

IV. Selection of Gears and Kinematic Actuation

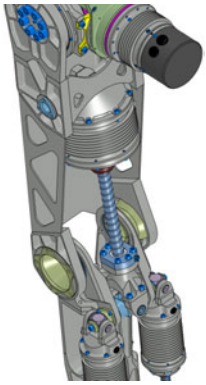
The demand for high torques at a minimum size and weight requires the use of reduction gears. The most important selection criteria besides the desired torque capacity are compact dimensions and low weight, as well as good dynamic performance and a sufficient torsional stiffness.

Harmonic Drive (HD) gears are used in many joints of the robot. The advantages of these gears are well known and include no-backlash and high reduction ratios within small space at low weight. The compact design of HD component sets allows a space-saving integration into the joint units [12]. All gears are custom lightweight versions with a T-shaped Circular Spline and Wave Generators, which are made from aluminum and modified for low weight and inertia. However, the HD gears exhibit high friction. Their frictional behavior is nonlinear and among other things depends on load, manufacturing and assembly tolerances. HD gears are backdrivable, however, efficiency decreases significantly in this case: A comparison of HD component sets has shown that the *no-load back driving torque* is approximately 23 % higher than the *no-load starting torque* which is required

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(a) Hip joint with intersecting axes



(b) Knee joint with linear actuator

Fig. 4. Hip joint and knee joint

to commence rotation of the Wave Generator with no load applied to the output flange.

On the other hand, ballscrews are perfectly suited for robotic applications due to high efficiency ($\eta \geq 90\,\%$), no backlash, no stick-slip and silent operation. As with HD gears, ballscrews are backdrivable, but here direct (motor is driving) and inverse (motor is driven) efficiency differ by less than 0.5 %. Therefore the reversal of movement (i.e. the conversion from linear to rotary motion) is more efficient than at HD gears. However, larger work ranges can hardly be realized with ballscrew drives and special care is necessary to avoid kinematic singularities. Although the drive itself is easier in design compared to a HD-based drive, ballscrew joints are much more complex than HD joints, because the whole surrounding construction is affected.

The second aspect besides gear selection is the actuation of multi-DoF joints by a serial or parallel drive configuration. The joint torques in parallel mechanisms are distributed almost equally on all actuators. These can be smaller in size which positively affects weight and power consumption of the robot. At the same time a parallel structure increases the stiffness of the joint. Parallel kinematics can be realized with HD gears or ballscrews. With 2 DoF each, the waist and ankle joints are perfectly suited for parallel mechanisms. Also it is desirable to have a parallel mechanism for the hip joint, however, the required work range of this 3-DoF joint makes the realization of an appropriate mechanism more difficult. As the only joint in the leg, the ankle joint is realized in a parallel configuration.

V. Mechanical Design of the Legs

Nearly all humanoid robots are designed with 6-DoF legs—3 DoF in the hip, one in the knee and two in the ankle. An additional link between forefoot and heel is introduced, equivalent to the human toes. Thus, heel lift-off is possible during terminal stance [10] and the swing leg can be in a more extended configuration. Area contact of the toe segment stabilizes the robot and facilitates forward roll across the forefoot which is expected to reduce the joint loads in hip and knee compared to a 6-DoF leg configuration.

A. Hip joint

The 3-DoF hip joint is of particular interest since it connects leg and pelvis and its overall stiffness considerably influences the walking performance. The hip joint shown in Fig. 4(a) is actuated by serial drives composing a spherical joint with axes intersecting in one point. Its compact design allows to keep the anthropometric proportions throughout the robot including the height of the torso. For better power distribution among the three hip drives, the yaw axis is inclined 15° to the vertical axis [15].

B. Knee joint

Torques and velocities of knee and hip pitch axis are similar, so the intention of a modular design originally implied the use of identical drives. However, using the hip joint module for the knee is problematic because its mass would unacceptably increase the thigh moment of inertia. In turn a large part of the enhanced hip joint output would be spent on accelerating a heavier thigh. Because of positive experience with ballscrews in the ankle joints of *Johnnie* [16] we went for the same actuation principle for the knees. By using a ballscrew drive in the knee, a large proportion of the drive's mass is shifted towards the hip, the thigh moment of inertia is kept low, and thus overall dynamics are improved. The high efficiency and backdrivability of the ballscrew is expected to be advantageous during the forward swing of the lower leg.

The actual mechanical configuration of this "muscle-like" mechanism is depicted in Fig. 4(b) and shown schematically on the left of Fig. 5. A better mass distribution in the hip-thigh area is achieved and *Lola*'s thigh moment of inertia is only marginally higher than *Johnnie*'s. Thus, the driving power of the knee could be enhanced without decreasing the hip joint's performance. The mechanism has nonlinear transfer behavior which is advantageous for typical gait patterns since the required torque markedly depends on the link position and has its maximum at around 50°. Fig. 5 shows the torque requirements in the knee for a walking speed of 5 [km/h] compared to the torque capacity of the drive and the velocity dependent capacity of the human knee. The trajectories have been calculated with a method based on nonlinear parameter optimization [17].

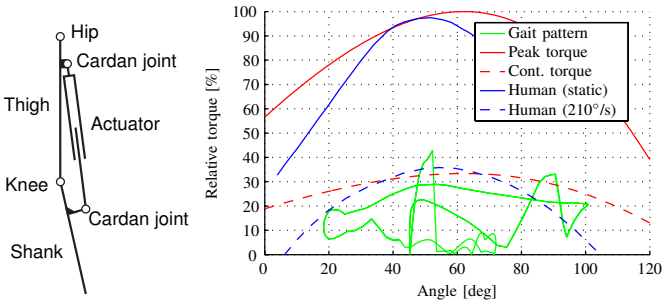


Fig. 5. Left: Mechanism employed for the the knee joint. Right: Torque and speed requirements of knee joint (human torque capacity taken from [10]).

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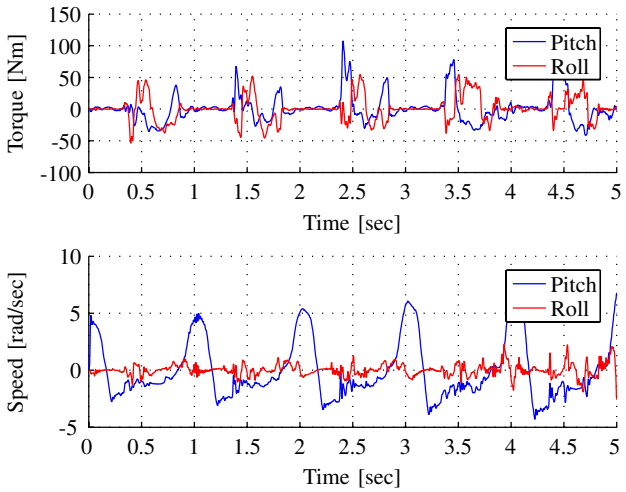


Fig. 6. Torque and speed requirements of the 2-DoF ankle joint at an approximate walking speed of 4 [km/h].

C. Ankle joint

Fig. 6 shows the torque and speed characteristics of the ankle joint at an approximate walking speed of 4 [km/h]. These data have been obtained from our multibody simulation of the robot. It has been verified experimentally with *Johnnie* and can thus be expected to be close to reality [14]. Obviously, both axes show clearly different torque-speed characteristics. The peak torque in the roll axis is 50 [Nm], whereas the pitch axis has a maximum torque of 120 [Nm]. The maximum joint velocities are 2 [rad/sec] in the roll axis and about 7 [rad/sec] in the pitch axis.

We compared several kinematic setups for both serial and parallel actuation of the ankle joint. The chosen concept is shown schematically in Fig. 7. The kinematics of the mechanism are designed so that only tension and compression loads act upon the drives. The drives are designed as length-variable steering rods, are identical in their mechanical design and kinematics, and thus reduce manufacturing effort and part diversity. Furthermore, the ballscrews permit precise control of the foot torques that act on the ground as they are free from starting torque and have very low rolling friction.

As an exemple, the procedure of calculating the relative

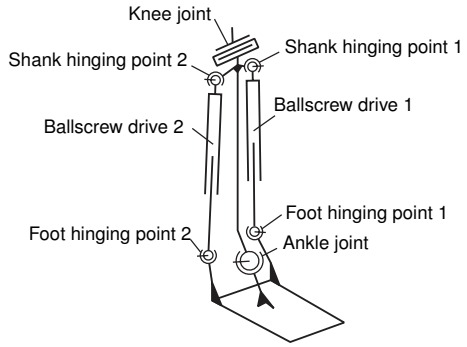


Fig. 7. Kinematic actuation principle of ankle joint

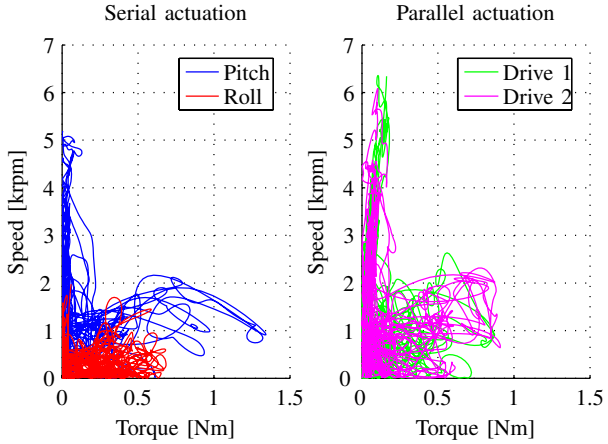


Fig. 8. Torque speed diagrams for the motors of the ankle joint in a serial and a parallel configuration.

Jacobian of rotation for a single ballscrew drive in the ankle joint is given (cf. Fig. 7). With the pitch and roll angles φ_p, φ_r the distance from the upper to the lower hinging point of the drive is

$$l_{BS} = \|\boldsymbol{r}_{ok} + \boldsymbol{A}_{UF}(\varphi_p, \varphi_r)\boldsymbol{r}_{kb} - \boldsymbol{r}_{oa}\| \ . \tag{1}$$

Here, $\boldsymbol{r}_{ok}$ denotes the vector from the origin of the shank segment to the ankle joint, $\boldsymbol{A}_{UF}$ is the transformation matrix from the shank into the foot coordinate system, $\boldsymbol{r}_{kb}$ is the vector from the ankle joint to the foot hinging point, and $\boldsymbol{r}_{oa}$ denotes the vector from origin of the shank segment to the the shank hinging point. For a given length l_{BS} the corresponding ballscrew rotation is

$$\varphi_{BS} = \frac{2\pi l_{BS}}{P_h} \ , \tag{2}$$

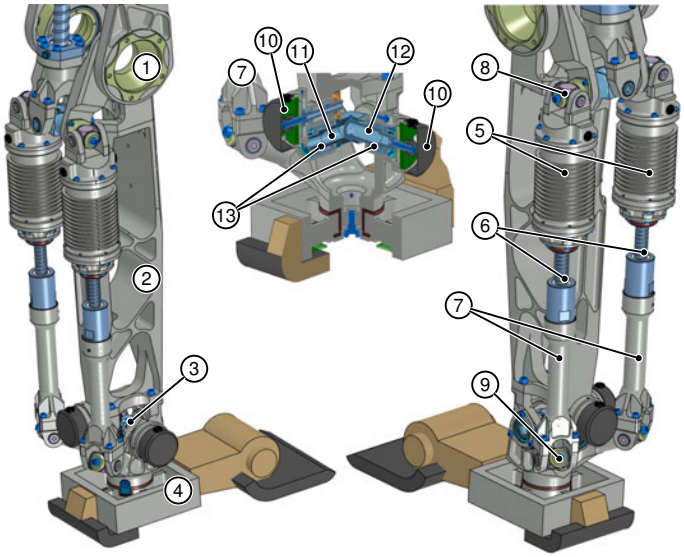
where P_h denotes the pitch of the ballscrew. The relative Jacobian can be calculated from (1) in a straightforward manner:

$$\frac{\partial \dot{\varphi}_{BS}}{\partial(\dot{\varphi}_r, \dot{\varphi}_p)^T} = \frac{2\pi}{P_h}\frac{\partial \dot{l}_{BS}}{\partial(\dot{\varphi}_r, \dot{\varphi}_p)^T} \tag{3}$$

If the ankle joint were built in a serial configuration, a rather powerful drive would be necessary for the pitch axis and a less powerful one for the roll axis. On the other hand, in the case of a parallel configuration both drives can be identical and smaller in size because the loads are distributed almost equally. The resulting torque speed diagrams of the ankle joint motors are shown in Fig. 8 for serial and parallel drives. The serial mechanism is constructed with Harmonic Drive gears, the parallel one with ballscrews. To be able to compare them, the reduction ratio of the Harmonic Drive gears is the average transmission ratio of the parallel mechanism ($N = 80$). By employing parallel drives, the peak torque is reduced by approx. 35 %, whereas the top speed increases by 20 %. However, these high velocities only occur at very low torques during the swing phase.

The realization of the ankle joint of *Lola* is shown in Fig. 9. The foot (4) is connected to the shank (2) via the ankle joint

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1	Knee joint	5	Drive	10	Angular sensor
2	Shank	6	Ballscrew	11	Roll axis
3	Ankle joint	7	Connection rod	12	Pitch axis
4	Foot	8,9	Cardan joint	13	Needle roller bearing

Fig. 9. Parallel mechanism in the ankle joint of *Lola*

(3) which is designed as a Cardan joint. Two linear drives consisting of a motor (5), ballscrew (6) and connection rod (7), actuate the foot. The connection rod (7) on the righthand side of the figure is transparent to illustrate the working principle of the drives: The ballscrew (6) plunges in the connection rod (7) so that the whole drive acts as a length variable steering rod. The motor (5) is placed as close as possible to the knee joint on order to minimize the inertia of the shank. The hinging points (8) coincide with the ones of the knee joint in order to improve load spread and distribution. Both hinging points (8,9) of motor and ballscrew nut are designed as cardan joints to keep the screw shaft free from radial loads. This simplifies the mechanical design since there are no linear bearings necessary to keep the ballscrew free from radial loads. The roll (11) and pitch (12) axis are designed as Cardan joint with intersecting axes. The cross unit (11,12) is made from rolling bearing steel and used as raceway for the needle roller bearings (13). This results in a very compact and stiff setup. The incremental encoder for motor control is mounted on the ballscrew (6), whereas the absolute angular encoders (10) are directly mounted on the joint axis.

VI. Conclusion

This paper presented the leg design of the 22-DoF humanoid walking robot *Lola*. The robot is characterized by its 7-DoF legs with actively driven toe joints. The joint drives show a modular, multi-sensory design with high performance AC servomotors. Harmonic Drive gears are used in many joints, but ballscrews are used in the waist, knee and ankle joints. In the knee joint they allow to shift the motor close to the hip which positively affects the mass distribution of the legs. In the ankle joints the superior frictional behavior of ballscrews permits precise control of the foot torques that act on the ground. Besides the modular joint concept, the informational structure of the robot as "intelligent sensor-actuator network" with *SERCOS III* as real-time communication system accounts for the reduction of overall complexity. *Lola*'s pelvis, thigh and shank are parts with complex loads and strict geometric constraints. They have been designed with topology optimization in order to balance the demands for high stiffness and strength with low weight.

The basic design of the robot's kinematic structure, modular joint design, sensor system and electronics architecture has been finished and a significant portion of mechanical parts has already been manufactured. Upon completion, optimal control design and walking experiments will be performed.

Acknowledgments

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